Arquitectura de Robots

Robot kinematics

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FIME
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA



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Robot kinematics

1 Robot kinematics

Robot kinematics studies the motion of the robot relative to a reference system.

Kinematics is interested in the analytical description of the spatial motion of the robot as a function of time, and in particular in **the relationships between the position and orientation** of the robot's end **with the values taken by its joint coordinates**.

Direct kinematic

Direct kinematic problem, and consists of determining the position and orientation of the final end of the robot, with respect to a coordinate system that is taken as a reference, **knowing the values of the joints and the geometric parameters of the robot elements.**

Inverse kinematic

Inverse kinematic problem, solves the configuration that the robot must adopt for a known position and orientation of the end.

Relationship between direct and inverse kinematics.

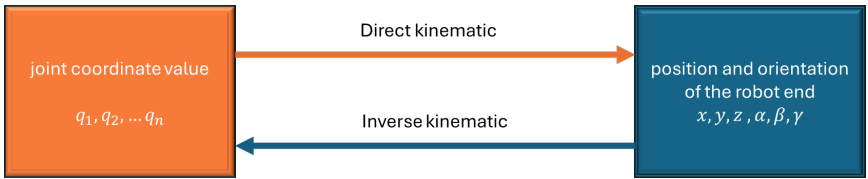


Figura 1: Relationship between direct and inverse kinematics

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- Robot kinematics
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Section 2.1

Kinematic geometric analysis

Kinematic geometric analysis

2 degree freedom manipulator (RP type).
Find the point P(x,y) of the robot

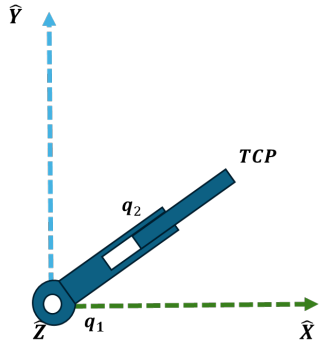


Figura 2: Robot 2 DoF RP type

Solution

2 Kinematic geometric analysis

Recognize the robot's articulations

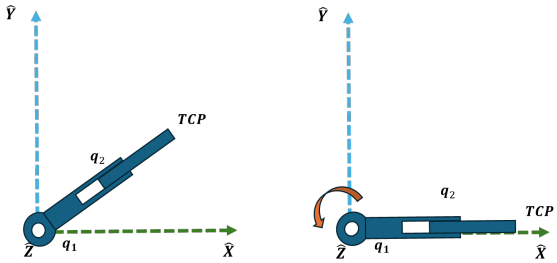


Figura 3: Robot 2 DoF RP type joint rotational

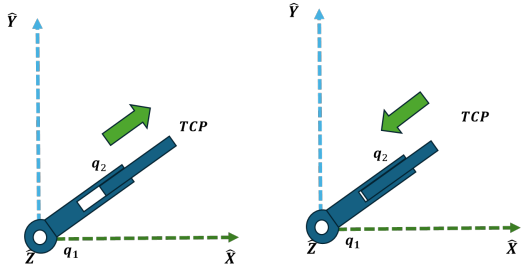


Figura 4: *Robot 2 DoF RP type join Prismatic.*

Solution

2 Kinematic geometric analysis

Determine the coordinates p_x, p_y, and p_z utilizing a geometric methodology.

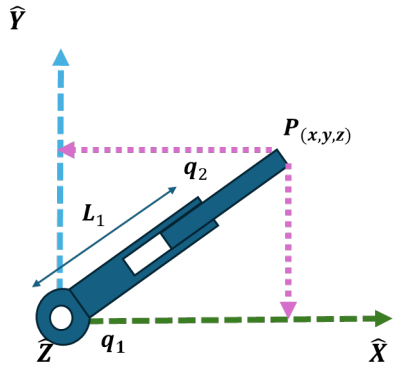


Figura 5: *Robot 2 DoF RP type .*

Solution

2 Kinematic geometric analysis

Build the transformation matrix to determine the final position and orientation of the Tool Center Point (TCP)

$$ {}^{0}_{2}T = \begin{bmatrix} {}^{A}_{B}R & {}^{A}P_{BORG} \\ 000 & 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R & P_{x,y,z} \\ 000 & 1 \end{bmatrix} $$

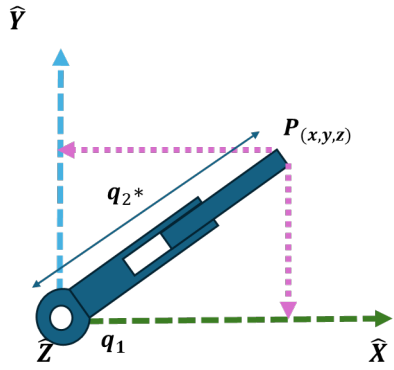


Figura 6: *Robot 2 DoF RP type .*

Solution

2 Kinematic geometric analysis

The variable q_2^* can be determined as the aggregate of the link L_1 and the displacement of the prismatic joint q_2.

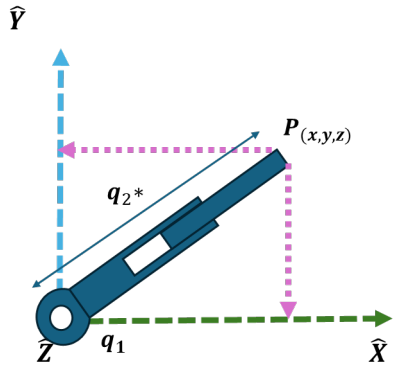


Figura 7: *Robot 2 DoF RP type .*

Solution

2 Kinematic geometric analysis

To determine m

$$\cos\theta = \frac{adj.}{hyp} = \frac{m}{q_2^*}$$

where $\theta = q_1$

$$m = q_2^* \cos q_1$$

if $m = p_x$

$$p_x = q_2^* \cos q_1$$

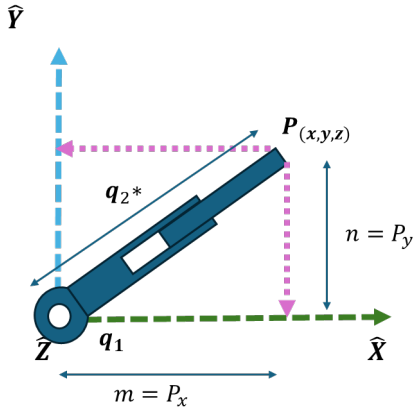


Figura 8: *Robot 2 DoF RP type .*

Solution

2 Kinematic geometric analysis

To determine n

$$\sin\theta = \frac{opp.}{hyp} = \frac{n}{q_2*}$$

where $\theta = q_1$

$$m = q_2^* \sin q_1$$

if $n = p_y$

$$p_y = q_2^* \sin q_1$$

The $p_z = 0$ component is zero due to the absence of vector components in the z-direction.

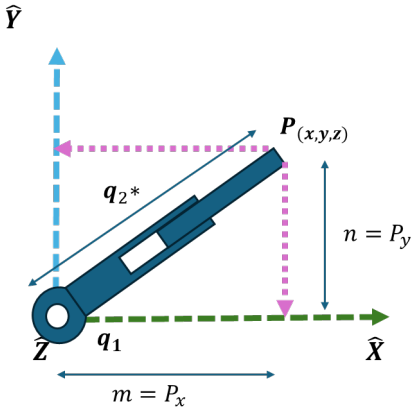


Figura 9: *Robot 2 DoF RP type .*

Solution

2 Kinematic geometric analysis

By substituting the position coordinates p_x, p_y, and p_z into the homogeneous transformation matrix, we obtain the following representation:

$$
{}^{0}_{2}T = \begin{bmatrix} {}^{A}_{B}R & P_{x,y,z} \\ 000 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & q_2^* \cos q_1 \\ r_{21} & r_{22} & r_{23} & q_2^* \sin q_1 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

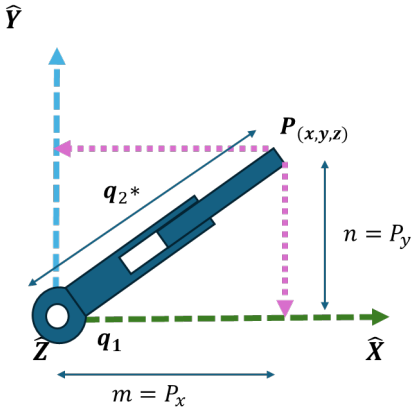


Figura 10: *Robot 2 DoF RP type .*

Solution

2 Kinematic geometric analysis

Now we get the TCP frame rotation based on the robot. The sole rotational component is the variable q_1, which is rotated about the z-axis.

$$\mathbf{R}_Z = \begin{bmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

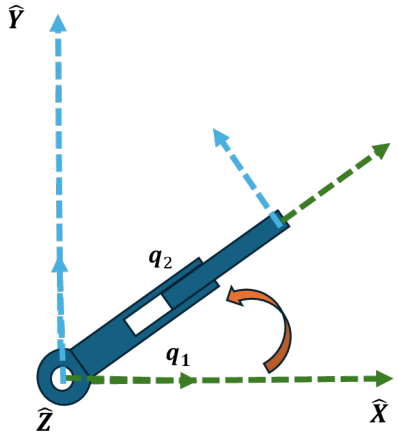


Figura 11: *Robot 2 DoF RP type* .

Solution

2 Kinematic geometric analysis

By substituting the rotational component, we derive the transformation matrix for the 2-degree-of-freedom (2DoF) robotic system.

$$
{}^{0}_{2}T = \begin{bmatrix} \cos q_1 & -\sin q_1 & 0 & q_2^* \cos q_1 \\ \sin q_1 & \cos q_1 & 0 & q_2^* \sin q_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

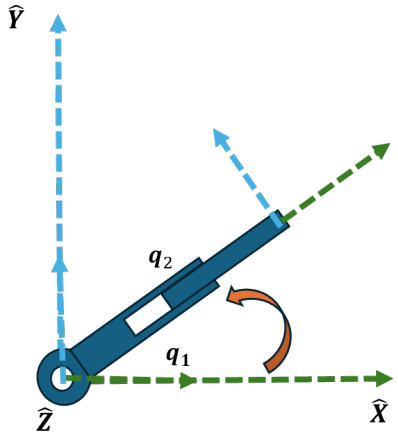


Figura 12: *Robot 2 DoF RP type* .

Section 2.2

Example 2

Kinematic geometric analysis 2 DoF RR type

2 degree freedom manipulator (RR type).
Find the point P(x,y) of the robot

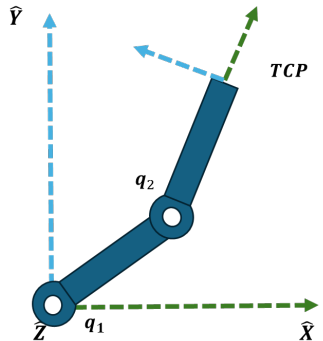


Figura 13: Robot 2 DoF RR type

Solution

2 Kinematic geometric analysis

Recognize the robot's articulations and links

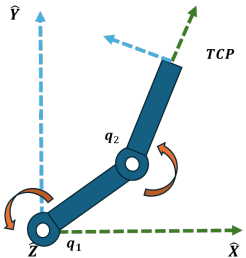


Figura 14: Robot 2 DoF RR type joint rotational

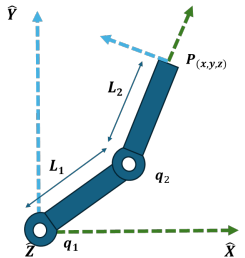


Figura 15: *Robot 2 DoF RR type join Prismatic.*

Solution

2 Kinematic geometric analysis

To determine m

$$\cos\theta = \frac{adj.}{hyp} = \frac{m}{L_1}$$

where $\theta = q_1$

$$m = L_1 \cos q_1$$

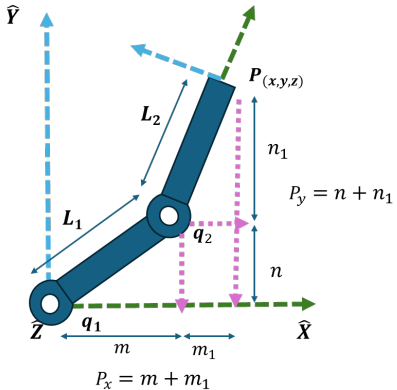


Figura 16: *Robot 2 DoF RR type .*

Solution

2 Kinematic geometric analysis

To determine m_1

$$\cos\theta = \frac{adj.}{hyp} = \frac{m_1}{L_2}$$

where $\theta = q_1 + q_2$

$$m_1 = L_2 \cos(\theta_1 + \theta_2) = L_2 \cos(q_1 + q_2)$$

$$P_x = m + m_1$$

$$P_x = L_1 \cos q_1 + L_2 \cos(q_1 + q_2)$$

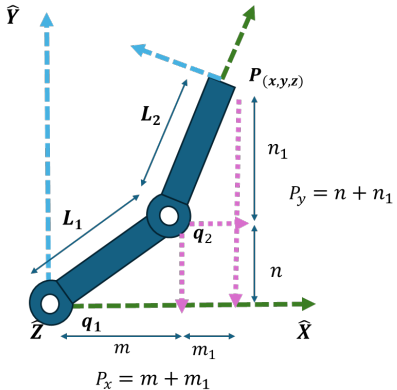


Figura 17: *Robot 2 DoF RR type .*

Solution

2 Kinematic geometric analysis

To determine n

$$\sin\theta = \frac{opp.}{hyp} = \frac{n}{L_1}$$

where $\theta = q_1$

$$n = L_1 \sin q_1$$

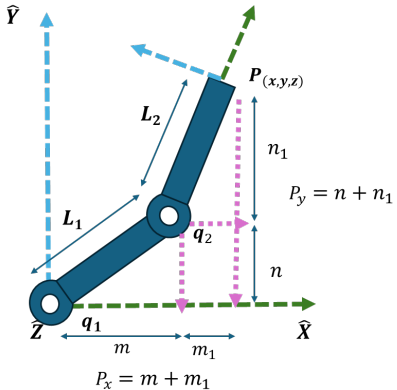


Figura 18: *Robot 2 DoF RR type .*

Solution

2 Kinematic geometric analysis

To determine n_1

$$\sin\theta = \frac{opp.}{hyp} = \frac{n_1}{L_2}$$

where $\theta = q_1 + q_2$

$$n_1 = L_2 \cos(\theta_1 + \theta_2) = L_2 \sin(q_1 + q_2)$$

$$P_y = n + n_1$$

$$P_y = L_1 \sin q_1 + L_2 \sin(q_1 + q_2)$$

The $p_z = 0$ component is zero due to the absence of vector components in the z-direction.

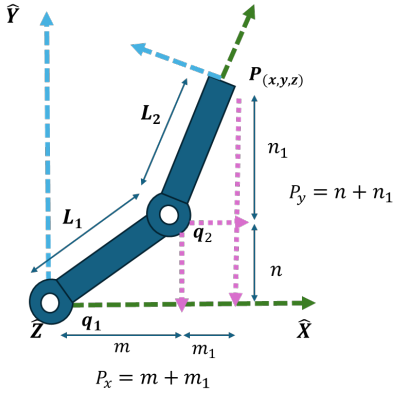


Figura 19: *Robot 2 DoF RR type .*

Solution

2 Kinematic geometric analysis

By substituting the position coordinates p_x, p_y, and p_z into the homogeneous transformation matrix, we obtain the following representation:

$$
{}^{0}_{2}T = \begin{bmatrix} {}^{A}_{B}R & P_{x,y,z} \\ 000 & 1 \end{bmatrix} =
$$

$$
\begin{bmatrix}
r_{11} & r_{12} & r_{13} & L_1\cos q_1 + L_2\cos(q_1+q_2) \\
r_{21} & r_{22} & r_{23} & L_1\sin q_1 + L_2\sin(q_1+q_2) \\
r_{31} & r_{32} & r_{33} & 0 \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

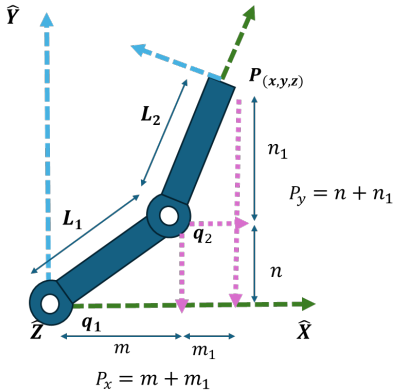


Figura 20: *Robot 2 DoF RR type .*

Solution

2 Kinematic geometric analysis

Now we get the TCP frame rotation based on the robot. The sole rotational component is the variable q_1 and q_2, which is rotated about the z-axis.

$$\mathbf{R}_{Z_1,Z_2} = \begin{bmatrix} \cos q_1 & -\sin q_1 & 0 \\ \sin q_1 & \cos q_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos q_2 & -\sin q_2 & 0 \\ \sin q_2 & \cos q_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

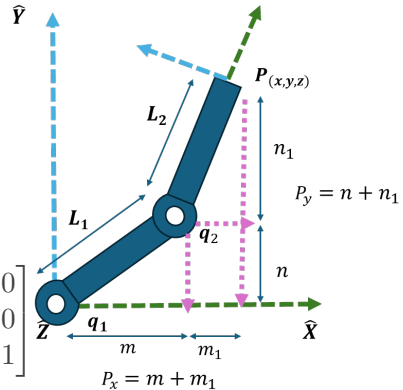


Figura 21: *Robot 2 DoF RR type .*

Solution

2 Kinematic geometric analysis

By performing the composition of the two rotational transformations about the Z-axis and utilizing the corresponding trigonometric identities, we derive the ensuing expression:

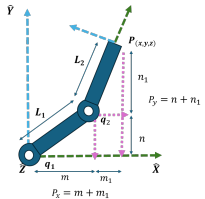


Figura 22: *Robot 2 DoF RR type .*

$$\mathbf{R}_{Z_1,Z_2} = \begin{bmatrix} \cos q_1 \cos q_2 - \sin q_1 \sin q_2 & -\cos q_1 \sin q_2 - \sin q_1 \cos q_2 & 0 \\ \sin q_1 \cos q_2 + \cos q_1 \sin q_2 & -\sin q_1 \sin q_2 + \cos q_1 \cos q_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution

2 Kinematic geometric analysis

Employing the appropriate trigonometric relations

$$\mathbf{R}_{Z_1,Z_2} = \begin{bmatrix} \cos q_1 \cos q_2 - \sin q_1 \sin q_2 & -\cos q_1 \sin q_2 - \sin q_1 \cos q_2 & 0 \\ \sin q_1 \cos q_2 + \cos q_1 \sin q_2 & -\sin q_1 \sin q_2 + \cos q_1 \cos q_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$$

$$\sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B)$$

$$\mathbf{R}_{Z_1,Z_2} = \begin{bmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & 0 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution

2 Kinematic geometric analysis

By substituting the rotational component, we derive the transformation matrix for the 2-degree-of-freedom (2DoF) robotic system.

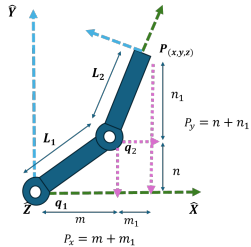


Figura 23: *Robot 2 DoF RR type .*

$$
{}^{0}_{2}T = \begin{bmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & 0 & L_1\cos q_1 + L_2\cos(q_1+q_2) \\ \sin(q_1+q_2) & \cos(q_1+q_2) & 0 & L_1\sin q_1 + L_2\sin(q_1+q_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

Section 2.3

Example 3

Kinematic geometric analysis 3 DoF RRR type

2 Kinematic geometric analysis

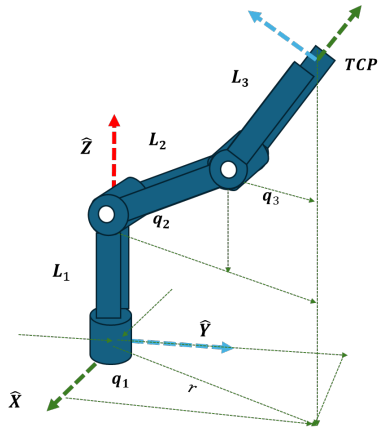


Figura 24: *Robot 3 DoF RRR type .*

Kinematic geometric analysis 3 DoF RRR type

Parameters and unknowns are established
Constants :

$$L_1, L_2, L_3$$

Variables :

$$q_1, q_2, q_3, r$$

The variable r is contingent upon the value of q_2, q_3. (does not constitute an articulation mechanism within robotic systems)

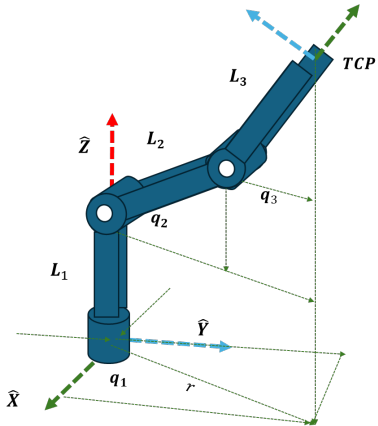


Figura 25: *Robot 3 DoF RRR type*

3 DoF RRR type solution

2 Kinematic geometric analysis

Determine the Px, Py, and Pz components of the position vector from the robot base to the Tool Center Point (TCP).

$$P_x = r\cos(q_1)$$

$$P_y = r sin(q_1)$$

$$P_z = L_1 + n + n_1$$

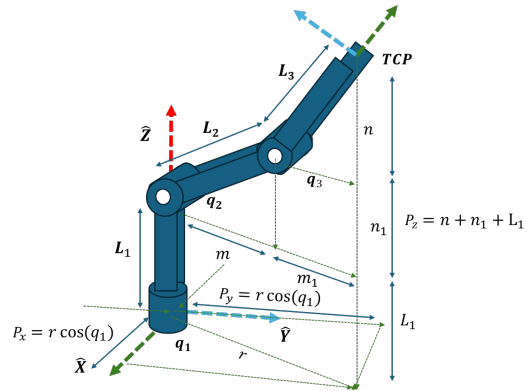


Figura 26: *Robot 3 DoF RRR type*

3 DoF RRR type solution

2 Kinematic geometric analysis

Calculate r

$$r = m + m_1$$

r represents the orthogonal projection of vectors m and m_1 onto the xy-coordinate plane.

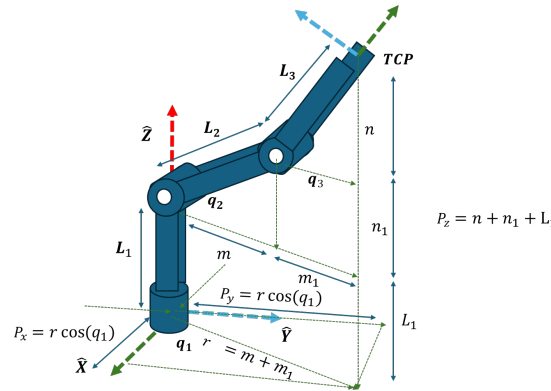


Figura 27: *Robot 3 DoF RRR type*

3 DoF RRR type solution

2 Kinematic geometric analysis

$$m = L_2 cos(q_2)$$

$$m_1 = L_3 cos(q_2 + q_3)$$

$$n = L_2 sin(q_2)$$

$$n_1 = L_3 sin(q_2 + q_3)$$

$$r = m + m_1 = L_2 cos(q_2) + L_3 cos(q_2 + q_3)$$

$$P_z = L_1 + n + n_1 = L_1 + L_2 sin(q_2) + L_3 sin(q_2 + q_3)$$

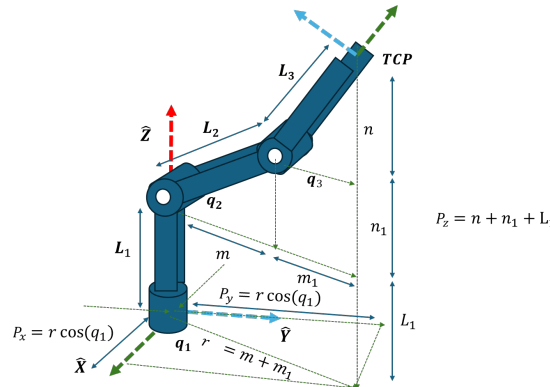


Figura 28: *Robot 3 DoF RRR type*

3 DoF RRR type solution

2 Kinematic geometric analysis

$$r = m + m_1 = L_2 cos(q_2) + L_3 cos(q_2 + q_3)$$

$$P_x = r cos(q_1)$$

$$P_x = (L_2 cos(q_2) + L_3 cos(q_2 + q_3))(cos(q_1))$$

$$P_y = r sin(q_1)$$

$$P_y = (L_2 cos(q_2) + L_3 cos(q_2 + q_3))(sin(q_1))$$

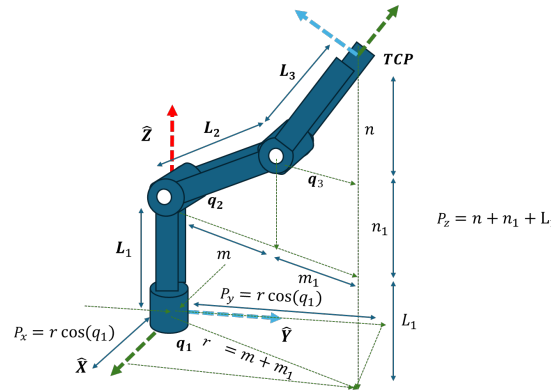


Figura 29: *Robot 3 DoF RRR type*

3 DoF RRR type solution

2 Kinematic geometric analysis

By substituting the position coordinates p_x, p_y, and p_z into the homogeneous transformation matrix, we obtain the following representation:

$$ {}^{0}_{2}T = \begin{bmatrix} {}^{A}_{B}R & P_{x,y,z} \\ 000 & 1 \end{bmatrix} = $$

$$ \begin{bmatrix} r_{11} & r_{12} & r_{13} & (L_2 cos(q_2) + L_3 cos(q_2 + q_3)) \cdot (cos(q_1)) \\ r_{21} & r_{22} & r_{23} & (L_2 cos(q_2) + L_3 cos(q_2 + q_3)) \cdot (sin(q_1)) \\ r_{31} & r_{32} & r_{33} & L_1 + L_2 sin(q_2) + L_3 sin(q_2 + q_3) \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Solution

2 Kinematic geometric analysis

Now we get the rotation of the robot from the base to the final orientation of the effector.

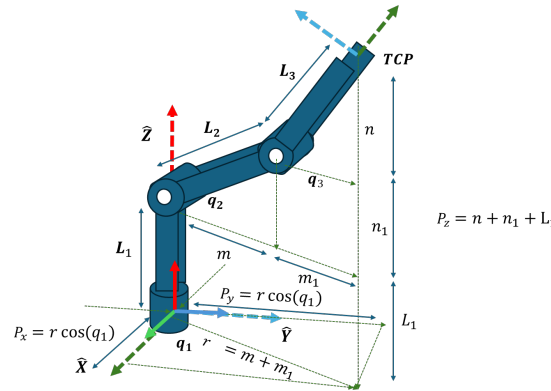


Figura 30: *Robot 3 DoF RRR type .*

Solution

2 Kinematic geometric analysis

Rotate about z to position x on the projection r

$$R_Z = (Z, q_1)$$

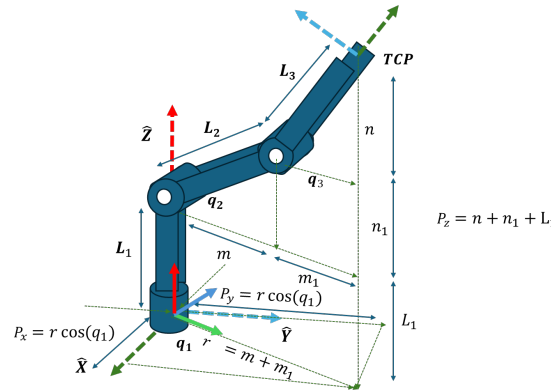


Figura 31: *Robot 3 DoF RRR type .*

Solution

2 Kinematic geometric analysis

Execute a rotation on the y-axis to position x along the trajectory L_2, this rotation is clockwise, so it is negative.

$$R_Y = (Y, -q_2)$$

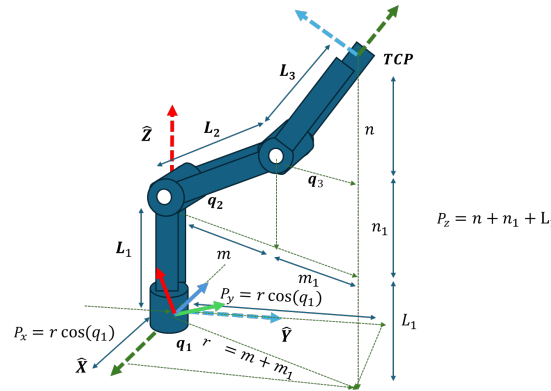


Figura 32: *Robot 3 DoF RRR type .*

Solution

2 Kinematic geometric analysis

Execute a additional rotation on the y-axis to position x along the trajectory L_3, this rotation is clockwise, so it is negative.

$$R_Y = (Y, -q_3)$$

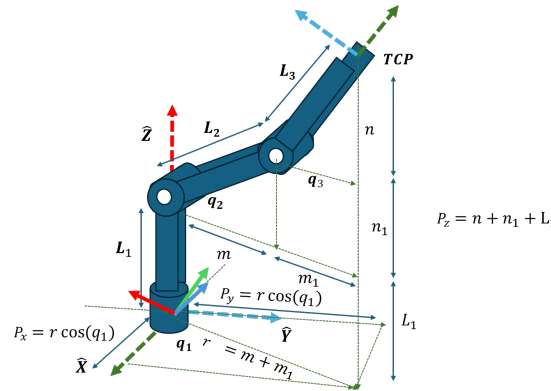


Figura 33: *Robot 3 DoF RRR type .*

Solution

2 Kinematic geometric analysis

Execute an auxiliary rotation about the y-axis to align the y-z coordinate axes with the terminal orientation of the TCP. This rotation is executed in a counterclockwise direction, and thus it is considered positive.

$$R_X = (x, 90)$$

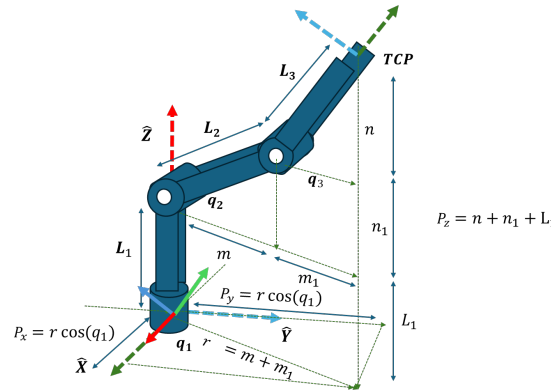


Figura 34: *Robot 3 DoF RRR type .*

Solution

2 Kinematic geometric analysis

Now we get the rotation of the robot from the base to the final orientation of the effector.

$$R(Z, q_1) \cdot R(Y, -q_2) \cdot R(Y, -q_3) \cdot R(X, 90)$$

$$\mathbf{R}_{Z,Y_1,Y_2,X} = \begin{bmatrix} cq_1 & -sq_1 & 0 \\ sq_1 & cq_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c(-q_2 - q_3) & 0 & s(-q_2 - q_3) \\ 0 & 1 & 0 \\ -s(-q_2 - q_3) & 0 & c(-q_2 - q_3) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c(90) & -s(90) \\ 0 & s(90) & c(90) \end{bmatrix}$$

$$\cos(-q_2 - q_3) = \cos(-q_2)\cos(-q_3) - \sin(-q_2)\sin(-q_3)$$

$$\sin(-q_2 - q_3) = \sin(-q_2)\cos(-q_3) + \cos(-q_2)\sin(-q_3)$$

Solution

2 Kinematic geometric analysis

Now we get the rotation of the robot from the base to the final orientation of the effector.

$$R(Z, q_1) \cdot R(Y, -q_2) \cdot R(Y, -q_3) \cdot R(X, 90)$$

$$\mathbf{R}_{Z,Y_1,Y_2,X} = \begin{bmatrix} cq_1 \cdot c(-q_2 - q_3) & cq_1 \cdot s(-q_2 - q_3) & sq_1 \\ sq_1 \cdot c(-q_2 - q_3) & sq_1 \cdot s(-q_2 - q_3) & -cq_1 \\ -s(-q_2 - q_3) & c(-q_2 - q_3) & 0 \end{bmatrix}$$

$$\cos(-q_2 - q_3) = \cos(-q_2)\cos(-q_3) - \sin(-q_2)\sin(-q_3)$$

$$\sin(-q_2 - q_3) = \sin(-q_2)\cos(-q_3) + \cos(-q_2)\sin(-q_3)$$

Solution

2 Kinematic geometric analysis

By substituting the rotational component, we derive the transformation matrix for the 3-degree-of-freedom (3DoF) robotic system.

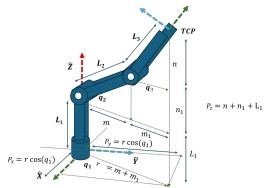


Figura 35: *Robot 3 DoF RRR type .*

$$
{}^{0}_{3}T = \begin{bmatrix}
cq_1 \cdot c(-q_2 - q_3) & cq_1 \cdot s(-q_2 - q_3) & sq_1 & (L_2c(q_2) + L_3c(q_2 + q_3)) \cdot (c(q_1)) \\
sq_1 \cdot c(-q_2 - q_3) & sq_1 \cdot s(-q_2 - q_3) & -cq_1 & (L_2c(q_2) + L_3c(q_2 + q_3)) \cdot (s(q_1)) \\
-s(-q_2 - q_3) & c(-q_2 - q_3) & 0 & L_1 + L_2s(q_2) + L_3s(q_2 + q_3) \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

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Q&A

Thank you for your attention

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Craig, J. J. (2006). Introduction to robotics. Pearson Educacion..
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